Tectonic implications of magnetic fabrics and remanence in the Cooper Mountain pluton, North Cascade Mountains, Washington

Tammy C. Fawcett, Russell, F. Burmester, Bernard A. Housen, and Alexander Iriondo

Abstract: Documenting the timing and kinematics of deformation in orogens is critical to unraveling their history. Anisotropy of magnetic susceptibility defines the orientation of magnetic fabrics in the Eocene Cooper Mountain pluton in the North Cascade Mountains of Washington. The magnetic foliation typically has a steep dip and a northwest strike; the magnetic lineation plunges moderately to shallowly northwest or southeast. The remanent magnetization was measured to determine if the Cooper Mountain pluton has been tilted following emplacement. The remanence has two components. The characteristic remanence typically unblocks at 370 °C in most specimens, but at 580 °C in others. The two components are carried by pyrrhotite and magnetite. Mean directions of these components are indistinguishable from each other and from the North American expected Eocene direction. The paleomagnetic results and ~ 47 Ma ^{40}Ar–^{39}Ar total fusion ages from biotite suggest that there has been no remagnetization or significant reorientation of the pluton since emplacement. Therefore, the in situ magnetic fabrics from the pluton can be used to understand the kinematics. Discordance of the fabrics with the pluton margin and near concordance with regional structures suggests that they have a tectonic origin. Thus the Cooper Mountain pluton is syntectonic rather than posttectonic. The magmatic fabric is slightly oblique to the length of the Cascade orogen, which can be explained if it formed as a consequence of regional dextral shear during transpression.

Résumé : Il est essentiel de documenter les facteurs temps et la cinématique de la déformation dans les orogènes afin de bien comprendre leur histoire. L'anisotropie de la susceptibilité magnétique définit l'orientation des fabriques magnétiques dans le pluton du mont Cooper (Éocène) dans les montages North Cascade de Washington. La foliation magnétique a typiquement un pendage abrupt et une direction nord-ouest; la linéation magnétique plonge modérément à légèrement vers le nord-ouest ou le sud-est. La magnétisation rémanente a été mesurée afin de déterminer si le pluton du mont Cooper a basculé depuis sa mise en place. La rémanence a deux composantes. La rémanence caractéristique typique dont la température de déblocage se situe à 370 ºC dans la plupart des spécimens mais à 580 ºC dans d'autres. Les deux composantes se retrouvent dans la pyrrhotite et la magnétite. Les directions moyennes de ces composantes ne peuvent être distinguées les unes des autres, ni de la direction attendue pour l'Amérique du Nord, à l'Éocène. Les résultats paléomagnétiques et les âges de fusion totale de 47 Ma ^{40}Ar–^{39}Ar d'échantillons de biotite suggèrent qu'il n'y a pas eu de remagnétisation ni de réorientation significative du pluton depuis sa mise en place. Donc, les fabriques magnétiques in situ du pluton peuvent être utilisées pour comprendre la cinématique. La discordance des fabriques avec la bordure du pluton et la quasi-concordance avec les structures régionales suggèrent une origine tectonique. Ainsi, le pluton du mont Cooper est syn-tectonique plutôt que post-tectonique. La fabrique magmatique est légèrement oblique par rapport à la direction longitudinale de l'orogène Cascade, ce qui peut s'expliquer s'il s'est formé suite au cisaillement régional dextre durant la transpression.

[Traduit par la Rédaction]

Introduction

The Cordillera in the U.S. Pacific Northwest has a complicated Mesozoic and Cenozoic tectonic history. This includes assembly of allochthonous terranes and their translation, contraction, and extension. Working out the tectonic history requires knowing the timing and kinematics of deformation. One of the candidates for faults that accommodated translation within the Cordillera is the Ross Lake fault zone (RLFZ; Figs. 1*a*, 1*b*). The RLFZ bounds the North Cascades

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crystalline core on the east; the Straight Creek fault bounds it on the west (SCF; Fig. 1*a*). There was dextral strike-slip faulting along the Straight Creek fault (Misch 1966) during the early Tertiary and possibly similar displacements on the RLFZ (Misch 1966; Haugerud et al. 1991*a*). An alternative interpretation is that the RLFZ is a continuous crustal section between two large-scale folds in the Skagit Gneiss Complex and Jurassic–Cretaceous Methow Basin (Kriens and Wernicke 1990). In the strike-slip scenario, the RLFZ accommodated dextral movement until about 45 Ma, when it was truncated by undeformed, posttectonic plutons with no observable displacement at their margins (Haugerud et al. 1991*b*). The Eocene Cooper Mountain pluton (CMP; Figs. 1*b*, 1*c*; Barksdale 1975) is one of these posttectonic plutons. The CMP intruded the southernmost extension of the Ross Lake fault zone known as the Foggy Dew fault (FDF; Figs. 1*b*, 1*c*). The fabric of the CMP might indicate whether the CMP is actually posttectonic and, if not, reflect the regional strain that affected it.

Fabric analysis of plutons and recognition of the difference between solid-state and magmatic fabrics (e.g., Paterson et al. 1989) have proven useful in understanding regional tectonics and mechanisms of plutonism. Magmatic fabrics in plutons can reflect strain during pluton emplacement and regional tectonism (see reviews by Bouchez 1997 and Paterson et al. 1998). In addition, magmatic fabrics develop during pluton crystallization, so they can constrain the age of deformation if the pluton is dated (Paterson et al. 1989).

Common problems encountered in the study of visible fabrics in plutons are that fabrics are variably developed, such that they may be unrecognizable in portions of a pluton where they are only weakly developed, they are difficult to measure on natural surfaces, and their development is difficult to quantify. Measuring magnetic fabrics is advantageous because three-dimensional exposure is not required and even subtle anisotropy produced by weakly developed mineral preferred orientation can be easily detected magnetically. The procedures most commonly used measure the magnetic susceptibility or difference in susceptibility as a function of direction within a rock to determine bulk fabric (anisotropy of magnetic susceptibility (AMS); see e.g., Uyeda et al. 1963; Hrouda 1982). For all minerals except magnetite, single crystal AMS is related to the mineral's crystallographic axes (Nye 1985). In many rocks, AMS is controlled by an Fe-rich silicate, such as biotite (Hrouda 1982; Borradaile and Henry 1997), but if the mafic minerals share the fabric with other silicates, measurements of the AMS of a rock should correspond to the bulk crystallographic preferred orientation and thus coincide with the petrofabric (Borradaile and Henry 1997; McNulty et al. 2000).

Fabric analysis, primarily by AMS, of the CMP was undertaken with the expectation that results would either bear on the emplacement mechanism of the pluton or lead to a better understanding of the regional tectonic history. Paleomagnetism of the CMP was also studied to evaluate whether the body as a whole has been reoriented since cooling. Some older plutons of the Cascades show significant paleomagnetic discordance with respect to the direction expected for that part of North America, indicating that either tilt or large-scale translation affected the rocks (e.g., the Cretaceous Mt. Stuart batholith; Beck and Nosen 1972). If paleomagnetism indicates tilting or rotation of a pluton, then the AMS fabrics must be reoriented before being used to document regional strain.

Geologic setting

The CMP, located in Chelan and Okanogan Counties in the North Cascade mountains of Washington (Fig. 1), is a 48 Ma (K–Ar biotite age; Tabor et al. 1980) granitic to granodioritic pluton with an aerial extent of 300 km^2 (Barksdale 1975). The CMP was emplaced during the waning of the Late Cretaceous orogeny in the North Cascades (Haugerud et al. 1994). The Late Cretaceous orogeny began in the Cretaceous and extended into the Tertiary (Mattinson 1972; Tabor et al. 1980; Haugerud et al. 1994). The orogeny included a significant component of orogen-parallel dextral transpression (Brown and Talbot 1989) and possibly some southwest-directed thrusting (Paterson and Miller 1998). The crystalline core itself is divided into two distinct blocks by the northwest-striking Entiat fault (Fig. 1), with the CMP located in the eastern, Chelan block. By the early Tertiary, the orogeny had ceased in much of the North Cascades crystalline core. This study is focused on the western side of the CMP where it intrudes rocks of the Skagit Gneiss Complex (Haugerud et al. 1991*b*) and Twenty-five Mile Creek unit (Figs. 1*b*, 1*c*). There are nearly 2 km of relief in this area along glacially carved Lake Chelan.

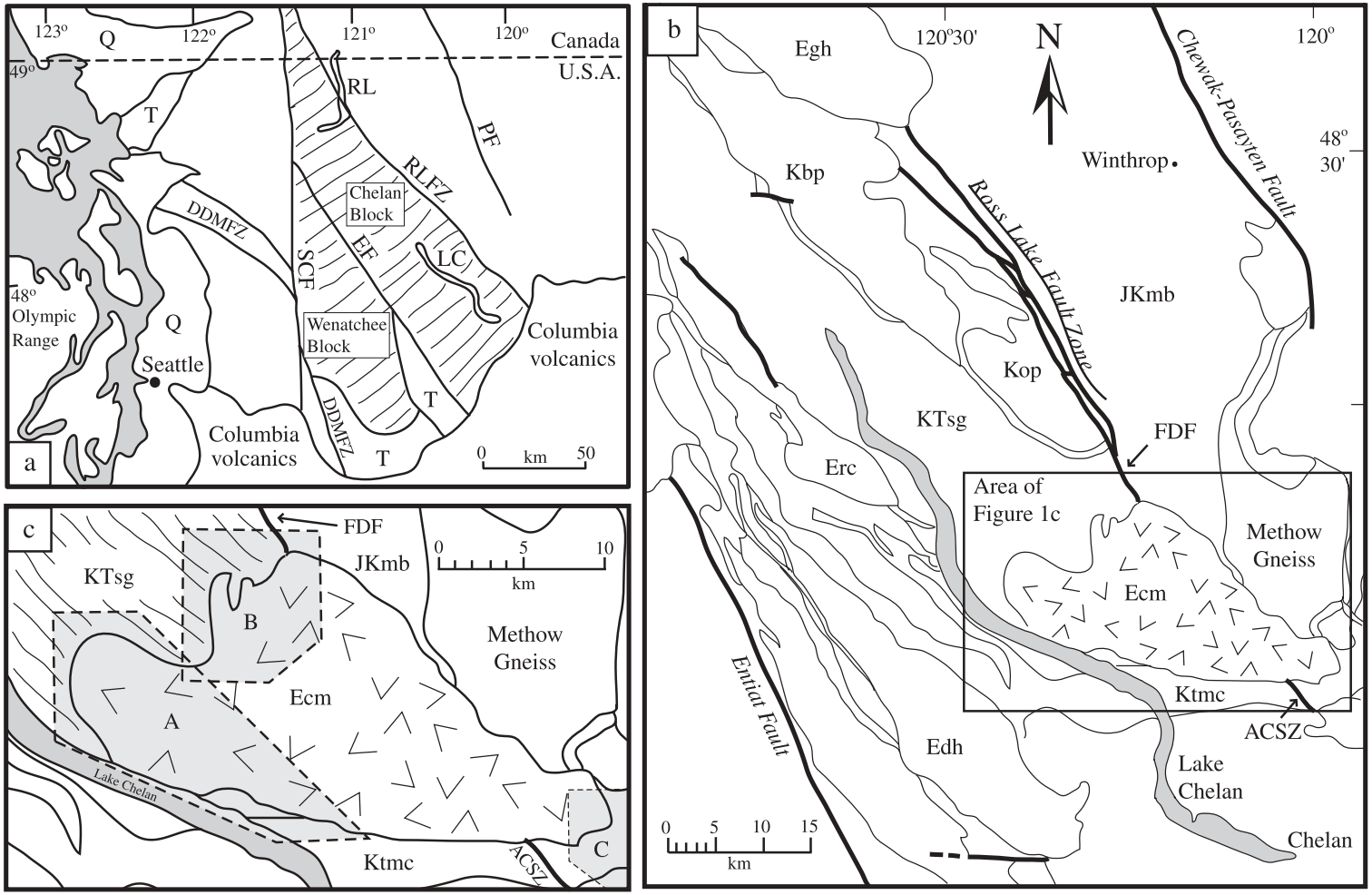
Previous structural analysis of the CMP was hampered by weak development of fabric. Wade (1988) looked for emplacement fabrics at the northern margin of the CMP where it truncates the Foggy Dew fault (Fig. 1*c*). Although fabric measurements were few because of weak development, Wade concluded that the foliations that parallel the CMP – Skagit Gneiss Complex boundary were caused by pluton emplacement. At the Foggy Dew fault contact he observed evidence of stoping, which led him to conclude that stoping was a probable mechanism for emplacement. Wade (1988) also described a small area of northwest-striking, moderately southwest-dipping foliation that does not fit margin-parallel flow. Raviola (1988) mapped the southeastern tip of the CMP (Fig. 1*c*) but was unable to map fabric because of weak fabric development.

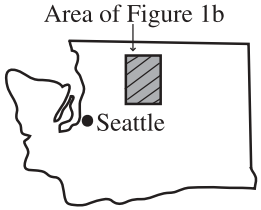
Emplacement of the CMP may have been by melt transported through fractures in the Skagit Gneiss Complex that formed from extension in the complex as it was being tilted to the southeast (Hopson and Mattinson 1999). Hopson and Mattinson (1999) provide evidence for tilting based on contact relations between the Skagit Gneiss and the CMP, a change in dike textures from deep to shallow crystallzation, and pressure relations all indicating increase in pressure from southeast to northwest in the Skagit Gneiss Complex. One goal of this paper is to investigate whether the apparent tilting of the gneiss has also affected the CMP.

We also found evidence that stoping was the emplacement mechanism of the CMP at the present level of exposure west of Wade's area. The pluton margin is a stockwork of dikes and sills and within the CMP, near the margin, there are blocks of Skagit Gneiss, whose foliation is discordant with that of the Skagit Gneiss country rock. Also, nowhere was there observed systematic deflection of the Skagit Gneiss around the CMP.

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Fig. 1. Location and geologic maps. (*a*) Geologic map of northwestern Washington emphasizing the regional faults (modified from Haugerud et al. 1994). (*b*) Location of the Cooper Mountain pluton (Ecm) in the eastern half of the North Cascades crystalline core, modified from Hopson and Mattinson (1994). (*c*) Approximate field areas for this and other studies (shaded regions). Figures 14*a* and 14*b* are the shaded area marked "A." LC, Lake Chelan; RL, Ross Lake.





Q	Quarternary sediments
T	Tertiary sedimentary units
Ecm	Cooper Mountain pluton
Egh	Golden Horn batholith
Edh	Duncan Hill pluton
Erc	Railroad Creek pluton
Kbp	Black Peak batholith
Kop	Oval Peak pluton

KTsg	Skagit Gneiss Complex
Ktmc	Twenty-five Mile Creek Unit
JKmb	Methow Basin

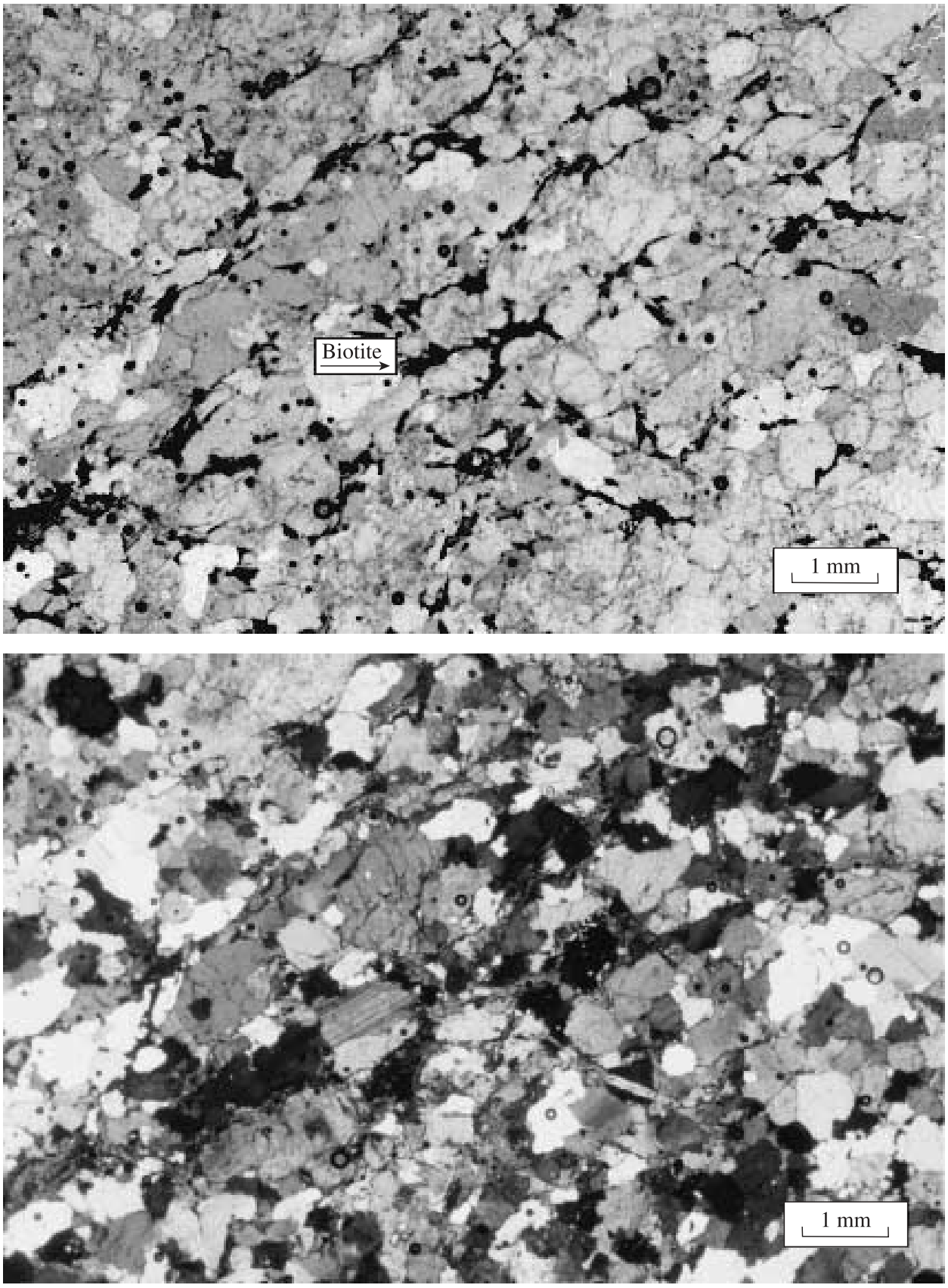
Approximate field area boundaries for Figure c:

A This study
B Wade (1988)
C Raviola (1988)

ACSZ	Antoine Coulee Shear Zone
DDMFZ	Darrington-Devils Mtn. Fault Zone
EF	Entiat fault
FDF	Foggy Dew fault
PF	Pasayten fault
RLFZ	Ross Lake Fault Zone
SCF	Straight Creek fault
———	Fault
———	Lithologic boundary

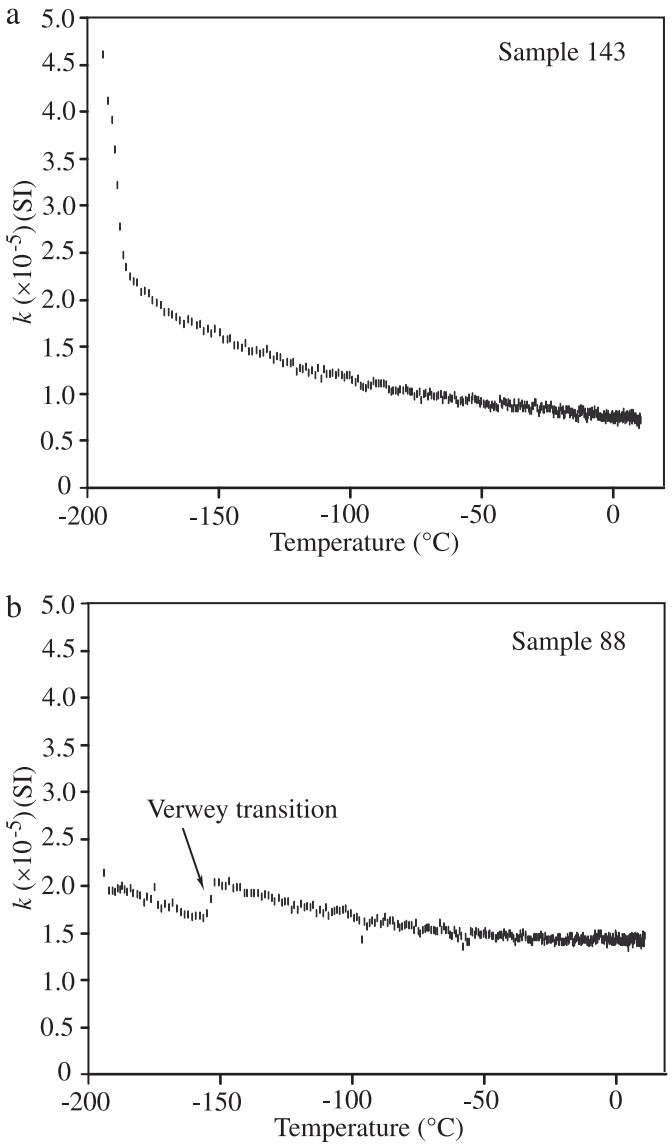
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Fig. 2. Photomicrographs of fine-grained granite sample 209b showing aligned biotites and lack of solid-state deformation. Specimen has k (magnetic susceptibility) = 6.29×10^{-5}, P (degree of anisotropy) = 1.212, T (ellipsoid shape parameter) = 0.344 (oblate susceptibility ellipsoid). Upper image in plane-polarized light, lower image in cross-polarized light.



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Fig. 3. Change of susceptibility during warming from –192 °C to room temperature. (*a*) Sample 143 displays a dominant paramagnetic behavior characterized by a smooth curve and higher susceptibility at lower temperatures (95% of susceptibility dominated by paramagnetics; K_p = 95%). (*b*) Sample 88 is dominated by ferromagnetic behavior shown by the small change in susceptibility with temperature and the Verwey transition at ~–150 °C (35% of susceptibility dominated by paramagnetics; K_p = 35%). Paramagnetic susceptibility, determined using the constant ferromagnetic susceptibility method of Hrouda (1994), is the total susceptibility accounted for by paramagnetic minerals.



Methods

Methods for this study addressed the petrologic and petrographic analysis of the pluton, magnetic mineralogy, remanence, and magnetic fabric. Fieldwork was concentrated along Lake Chelan (Fig. 1), logging roads, and trails where good exposures existed. The area was mapped to determine pluton contacts, petrographic variation, and structural fabric. One-hundred and twenty oriented hand samples, from 115 sites, were collected for AMS, paleomagnetic, and petrographic analyses (Appendix A, Table A1). Thirty-five of these were slabbed and stained to discriminate plagioclase from potassium-feldspar, then counted to determine modes. Thirteen oriented thin-sections were made of samples chosen to represent the range of recognized lithologies, AMS, and spatial variation throughout the field area.

In the lab, 109 oriented hand samples were drilled with a nonmagnetic bit to obtain one to five cores (typically 2) per block sample. Two to four specimens were cut from each core with a nonmagnetic diamond saw blade. Magnetic susceptibilities of these specimens were measured on a Kappabridge KLY-3 susceptometer. Site-mean AMS was calculated using the methods of Constable and Tauxe (1990). Specimens were stored in a magnetically shielded room, where the remanent magnetization of specimens from 57 sites was measured with a 2G-755 DC SQUID cryogenic magnetometer and demagnetized in a custom-built magnetically shielded oven and a D-Tech alternating field demagnetizer. Linear segments of demagnetization paths were visually identified on orthogonal vector endpoint diagrams (Zijderveld 1967) as components of magnetization. Directions of those components and their maximum angular deviation (MAD) were obtained with principal component analysis (Kirschvink 1980). Calculation of mean directions and statistics followed Fisher (1953).

After results from other phases of the investigation were complete, two of the freshest samples available were selected from the southwest and southeast limits of the field area to represent both of the recognized main phases of the CMP. The samples were crushed and biotite grains were separated at the geochronology lab at The University of British Columbia, Vancouver, B.C. Biotites from two granite–granodiorite samples from the Cooper Mountain pluton (137a and 235) were separated to perform $^{40}Ar/^{39}Ar$ geochronology (Fig. 12 and Table A2). Mineral separates (250–180 μm-size fraction) were produced using magnetic separation, heavy liquids, and hand picking techniques to a purity of > 99%. The samples were then washed in acetone, alcohol, and deionized water in an ultrasonic cleaner to remove dust and then re-sieved by hand using a 180-μm sieve.

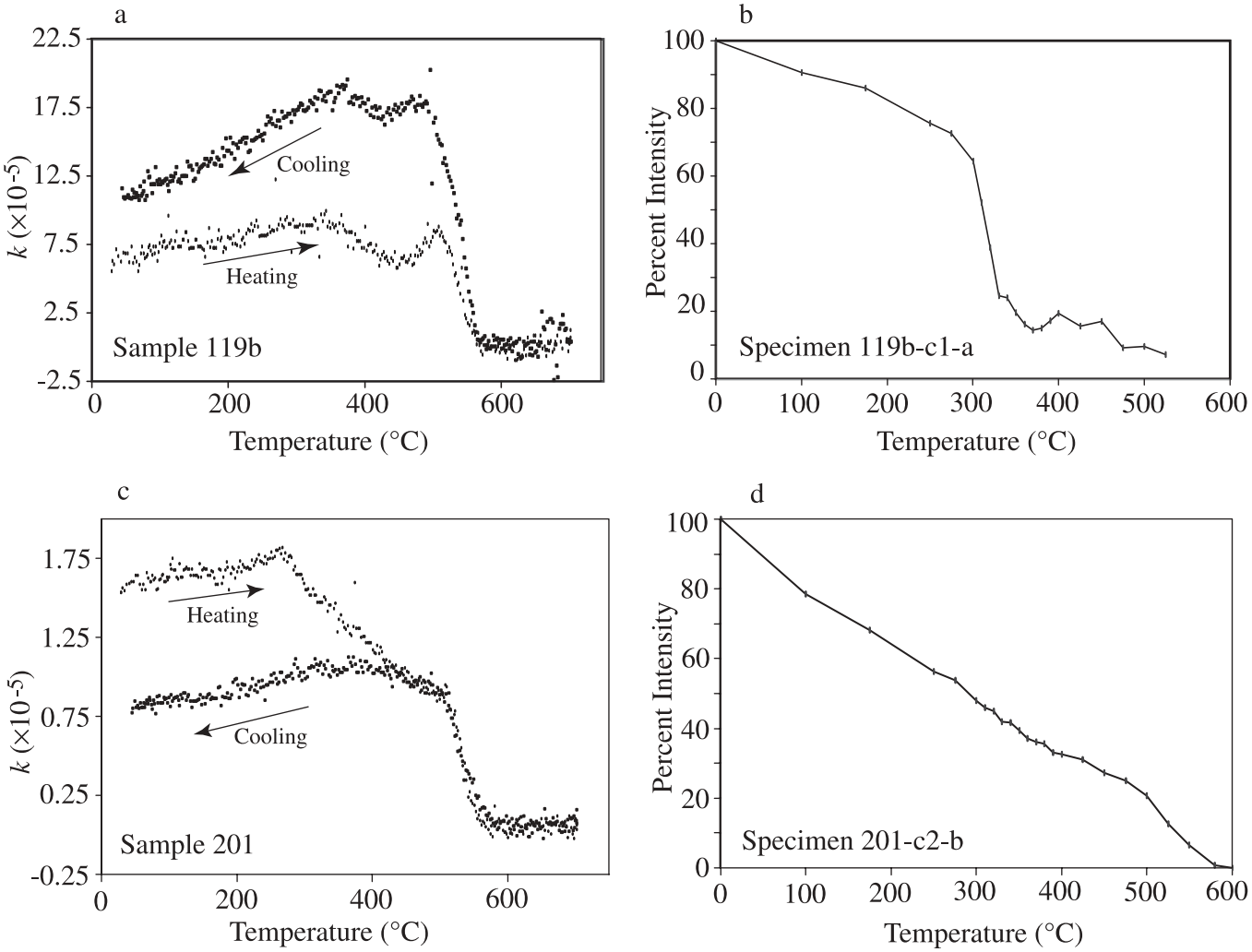
Biotite aliquots of ~ 20 mg (step heating) and 3 mg (total fusion) from both samples were packaged in copper capsules and sealed under vacuum in quartz tubes. The samples were irradiated for 20 h (irradiation package KD22) in the central thimble facility at the TRIGA reactor (GSTR) at the U.S. Geological Survey, Denver, Colorado. The monitor mineral used in the package was Fish Canyon Tuff sanidine (FCT-3) with an age of 27.79 Ma (Kunk et al. 1985; Cebula et al. 1986) relative to MMhb-1 with an age of 519.4 ± 2.5 Ma (Alexander et al. 1978; Dalrymple et al. 1981). The type of container and the geometry of samples and standards is similar to that described by Snee et al. (1988).

Both biotite samples were analyzed at the U.S. Geological Survey (USGS) Thermochronology Laboratory in Denver, Colorado on a VG Isotopes Ltd., Model 1200 B Mass Spectrometer fitted with an electron multiplier using the $^{40}Ar/^{39}Ar$ step-heating and total fusion methods of dating. For additional information on the analytical procedure see Kunk et al. (2001).

The argon isotopic data were reduced using an updated

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Fig. 4. High temperature thermomagnetic plots (*a*, *c*) show a dramatic drop in magnetic susceptibility at ~580 °C, which corresponds to magnetite's Curie temperature. Accompanying normalized remanence thermal demagnetization plots (*b*, *d*) show that specimens from the same samples have two different unblocking temperatures (320 and 580 °C).



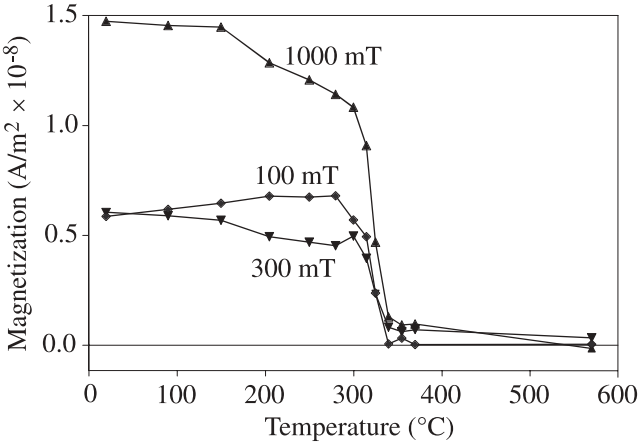
version of the computer program ArAr* (Haugerud and Kunk 1988). We used the decay constants recommended by Steiger and Jäger (1977). Table A2 shows $^{40}Ar/^{39}Ar$ step-heating and total fusion data for the biotites and includes the identification of individual steps, existence of plateau, and total gas ages. Total gas ages represent the age calculated from the addition of all of the measured argon peaks for all steps in a single sample. The total gas ages are roughly equivalent to conventional K/Ar ages. No analytical precision is calculated for total gas ages. Plateau ages are identified when three or more contiguous steps in the age spectrum agree in age, within the limits of analytical precision, and contain more than 50% of the ^{39}Ar released from the sample.

Results

Petrology

Texture in the CMP ranges from fine-grained (0.1–0.3 cm grains) (Fig. 2) to equigranular coarse-grained (0.1–0.6 cm grains) to porphyritic (0.1–0.7 cm groundmass) with phenocrysts of potassium-feldspar up to 2.0 cm in size. Modal compositions of the porphyritic and non-porphyritic phases are similar,

Fig. 5. Thermal demagnetization of a three axis isothermal remanent magnetization given specimen 66-c1-b following the method of Lowrie (1990). Fields used were 1000, 300, and 100 mT. Notice the drop in all magnetizations between 320 and 340 °C. See tecxt for discussion.



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with their compositions falling near the granite–granodiorite boundary (Streckeisen 1976). The porphyritic phase contains 29%–47% quartz, 33%–48% plagioclase, 9%–26% potassium-feldspar, and 2%–13% mafic minerals. The non-porphyritic rocks contain 30%–50% quartz, 30%–47% plagioclase, 13%–25% potassium-feldspar, and 1%–8% mafic minerals. The mafic constituents are biotite ± hornblende. Biotite occurs as thin books about the same size as but thinner than feldspars and irregular, interstitial flakes. Hornblende occurs as grains smaller and less abundant than biotite. Accessory minerals constitute < 1% of the mode of all rocks and include apatite, zircon, and opaques. Only ilmenite and magnetite were identified as opaque minerals using reflected light microscopy. See the following section for estimates of nature and abundance of magnetic phases.

Magnetic mineralogy

A variety of experiments showed that both single- and multidomain magnetite, as well as pyrrhotite, occur in the rocks studied (Fawcett 2001). Some these are illustrated here to show that biotite dominates the magnetic behavior of some samples despite the general contribution of magnetically ordered phases, and that pyrrhotite, not seen in thin section, is widespread.

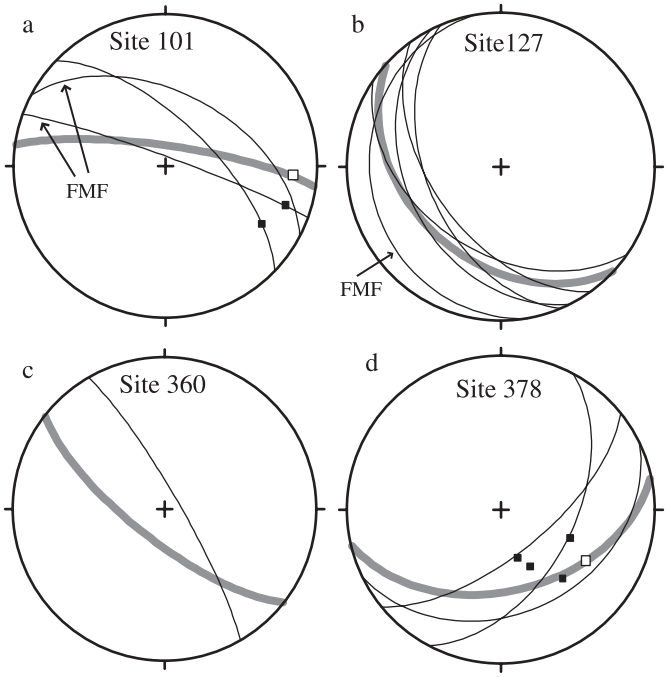
Presence of magnetite can be demonstrated by its change in susceptibility at low and high temperatures. The Kappabridge cryostat CL-3 was used to measure magnetic susceptibility during warming from –192 to 10 °C. Specimens from 16 sites, thought to span the range of bulk susceptibility observed, were analyzed. Eleven samples (69%) had smooth paramagnetic curves (Fig. 3*a*) and no evidence of magnetite contribution. Five specimens (31%) had a magnetite signature, consisting of an increase in susceptibility upon warming through the Verwey transistion (Fig. 3*b*), and 1 (6%) showed equal paramagnetic and magnetite contribution. The presence of magnetite in some samples was also demonstrated by an abrupt change in susceptibility at 580 °C (the Curie temperature of magnetite) and presistance of remance to 580 °C (Fig. 4*d*).

The presence of pyrrhotite also is demonstrated using the thermomagnetic curves and demagnetization of remanence. The inflections at about 300 °C in Figs. 4*a* and 4*c* and the observation that most specimens lose 90% of their intensity between thermal demagnetization steps of 300 °C and 350 °C (Fig. 4*b*) are consistant with the presence of pyrrhotite, with Curie temperature T_c = 320 °C. Occurrance of pyrrhotite is perhaps better demonstrated with demagnetization of an artificial remanence. We imparted isothermal remanence using an ASC impulse magnetizer to eight specimens following the method of Lowrie (1990). Magnetizing fields were 200 mT along the specimens' *Z* coordinate direction, 80 mT along *Y*, and 30 mT along *X*. These magnetizing fields were chosen to match the coercivity ranges of the expected magnetic minerals. Figure 5 shows that remanence along all three axes was unblocked around 340 °C, consistent with pyrrhotite being the dominant magnetic mineral. The fact that substantial remanence is retained in directions of all three magnetizing fields indicates that the coercivity spectrum of the pyrrhotite is broad, perhaps due to a wide range in grain size.

How the magnetic fabric should be interpreted depends on which minerals contribute to a specimen's susceptibility and its anisotropy. Using modes from Fawcett (2001) and susceptibilities of mineral species from Borradaile and Henry (1997), we estimated the contributions of diamagnetic (quartz and feldspars) and paramagnetic (biotite) minerals for each sample and subtracted their sum from the same sample's bulk susceptibility to estimate the contribution from magnetically ordered phases (e.g., magnetite, pyrrhotite). Since pyrrhotite appears to be widespread as a trace mineral, it may account for much of the background ordered phase contribution to the susceptibility. Magnetite is present as both large, multidomain grains that are visible in some samples and as single-domain grains (Fawcett 2001). One possible location for the latter is in biotite, where their contribution to susceptibility would already have been accounted for (the susceptibility used for biotite probably includes some ferromagnetic contribution; Borradaile and Henry 1997). The ordered phase contribution implies magnetic phases constitute about 0.005% of the rock by volume if all such phases are pyrrhotite, half that if coarse grained magnetite, or 10 times that if ilmenite. Although ilmenite was visible in some samples, it is paramagnetic at room temperature so does not carry the remanent magnetization found in these rocks. We consider its contribution to the susceptibility to be negligible.

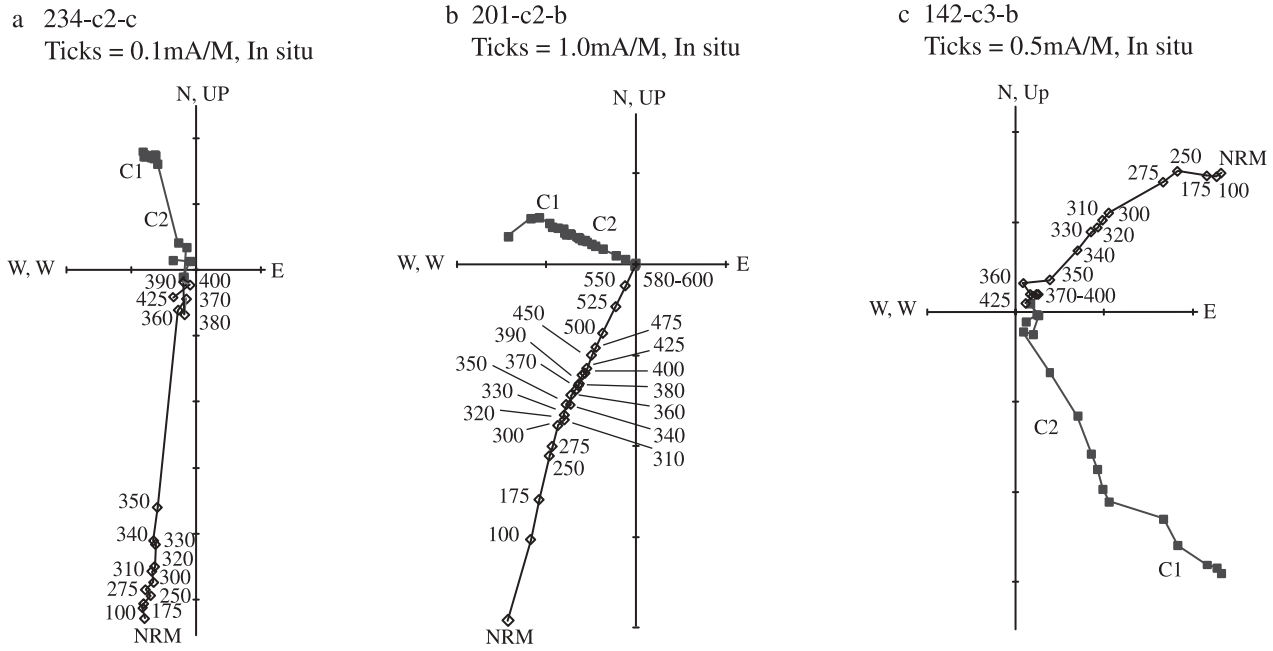
The calculation in the preceding paragraph shows that for most samples, biotite and ordered phases contribute approximately equally to the total susceptibility, which is about 10 times the negative contribution of the diamagnetic phases.

Fig. 6. Lower hemisphere projections of fabrics for the four specimens that displayed ferromagnetically controlled susceptibility in low-temperature thermomagnetic experiments. The AMS data for each specimen (thick, gray lines for foliation; open squares for lineation) are plotted along with magnetic fabric from adjacent paramagnetic sites and with fabric measured in the field (thin, black lines for foliation; solid squares for lineation). FMF, field measured foliation, others are AMS fabrics. Overall, the ferromagnetically and paramagnetically controlled fabrics are similar.



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Fig. 7. Orthogonal plots of vector end points for three paleomagnetic specimens. Demagnetization paths bend between 300° and 350°, defining two components labeled C1 and C2 on horizontal projection or map view (solid squares). Thermal demagnetization steps are labeled in °C on west–east vertical projections (open diamonds). NRM, natural remanent magnetism.



We think that this is an overestimate of the contribution from ordered phases based on the low-temperature experiment (Fig. 3) and other rock-magnetic evidence, such as high-field susceptibility (Fawcett 2001). Instead, we conclude that biotite is responsible for the bulk of the susceptibilty in the rocks. Whether biotite or a magnetic mineral such as pyrrhotite and (or) magnetite controlled the susceptibility anisotropy would depend on their relative intrinsic anisotropies and efficiencies of alignment. Both magnetite and pyrrhotite can have very high intrinsic anisotropies, depending upon their domain state and grain shapes. However, similarity of visible fabrics measured in the field and AMS fabrics of samples (Fig. 6) suggests that any contribution from the ordered phases to the total AMS at least does not greatly counteract the paramagnetic contribution.

Remanence

Thermal demagnetization of NRM showed two unblocking temperature ranges. Remanence in most (N = 48) specimens was unblocked by 370 °C; magnetization in all but one of the remaining specimens (N = 17) was unblocked by 580 °C. Several specimens had two remanence components (Fig. 7). Since the inflection point between the two components on the vector end-point plots was near the Curie temperature of pyrrhotite in all specimens, the component demagnetized at lower temperatures is referred to as the pyrrhotite component and the higher temperature component is referred to as the magnetite component.

Directions of both pyrrhotite and magnetite components are scattered, with a minority being upward (Figs. 8*a*, 8*b*; Table A3). This suggests that some of the remanence dates from a time of reverse polarity of the magnetic field. Some of the scatter might result from magnetizations recording transitional fields, or a mixture of the two polarities. To help reduce the contribution to scatter from demagnetization paths that result from simultaneous demagnetization of opposite polarity magnetizations, only line segments with MAD < 8° were used. Furthermore, to reduce bias that might arise from subjectively separating the polarities to calculate their mean directions, the *bootdi* method of Tauxe (1998) was used. This first employs a principle component analysis (Kirschvink 1980) approach to divide the data set into opposite modes, then it calculates Fisher means and statistics for each mode. Upward directions were too scattered to yield useful results. The mean of the downward mode of the pyrrhotite component from 42 specimens is D (declination) = 343.5°, I (inclination) = 73.2°, α_{95} (95% confidence level) = 6.7°, and k (precision parameter) = 12. The mean direction of the magnetite component from 16 specimens is D = 342.7°, I = 72.8°, α_{95} = 9.2°, and k = 17. The means of these two components are unlike the present day field direction (D = 20.9°, I = 71.7°) but similar to each other and the direction expected for the Eocene (D = 349.5°, I = 67.4°; calculated from North American pole of Diehl et al. 1983) (Fig. 8*c*).

Anisotropy of magnetic susceptibility

Average susceptibilities of 104 sites from which AMS was determined range from 1.72×10^{-5} to 6.64×10^{-4} SI, with an average bulk susceptibility for the study of 1.10×10^{-4} SI (Table A4). Only two specimens had anisotropies that were insignificant at the 95% confidence level judging from the F-statistics calculated (Tauxe 1998). After omitting those two specimens, site mean AMS and confidence limits were calculated using a bootstrap procedure (Constable and Tauxe 1990). The maximum, intermediate, and minimum axes of susceptibility (k_n) are denoted throughout the text as k_{max}, k_{int}, and k_{min}, respectively. Magnetic foliations are the planes normal to k_{min} defined by the k_{max}–k_{int} orientations, and magnetic lineations are the orientations of k_{max}. If all three eigenvectors and their eigenvalues are distinct (Figs. 9*a*,

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Fig. 8. Paleomagnetic directions of pyrrhotite and magnetite components. (*a*) and (*b*) are equal-area projections for individual specimens. Closed circles are downward directions, opened circles are upward directions. (*c*) Pyrrhotite and magnetite mean directions (black squares) plotted with their α_{95} confidence limits. Also plotted are the present-day field direction (star) and the North American expected Eocene direction (gray diamond, calculated for the location from pole of Diehl et al. (1983); α_{95} circle of confidence for the expected Eocene direction is as small as the symbol drawn).

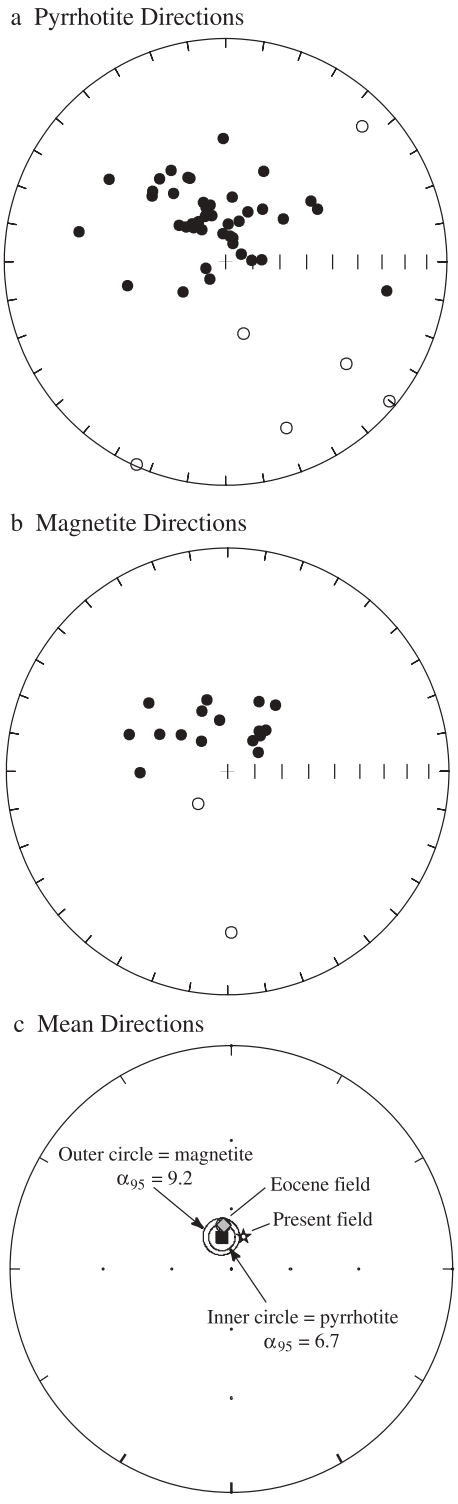
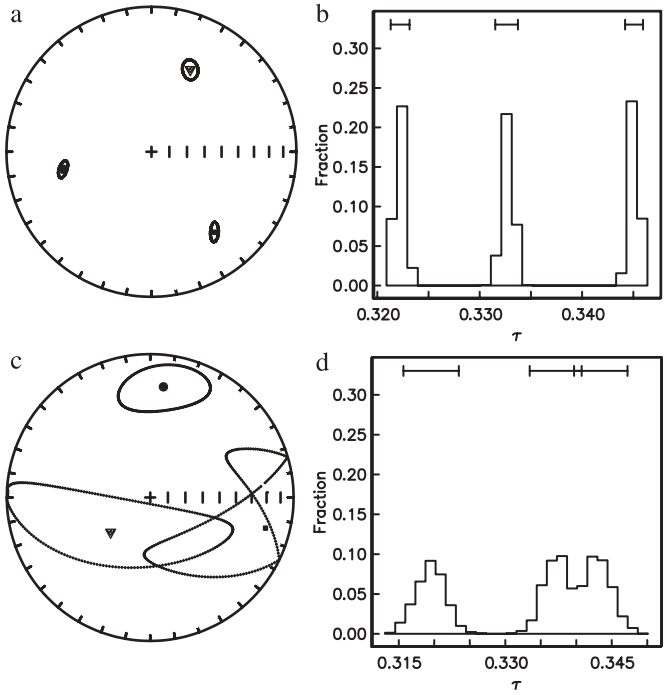


Fig. 9. Bootstrap statistical analysis lower hemisphere equal-area projections and histograms. (*a*) and (*c*): squares, k_{max} mean eigenvector; triangles, k_{int} mean eigenvector; circles, k_{min} mean eigenvector; bootstrapped error ellipses surround each axis. (*b*) and (*d*): Fraction, fraction of bulk susceptibility; τ, eigenvalue; bootstrapped 95% confidence bars above each histogram.

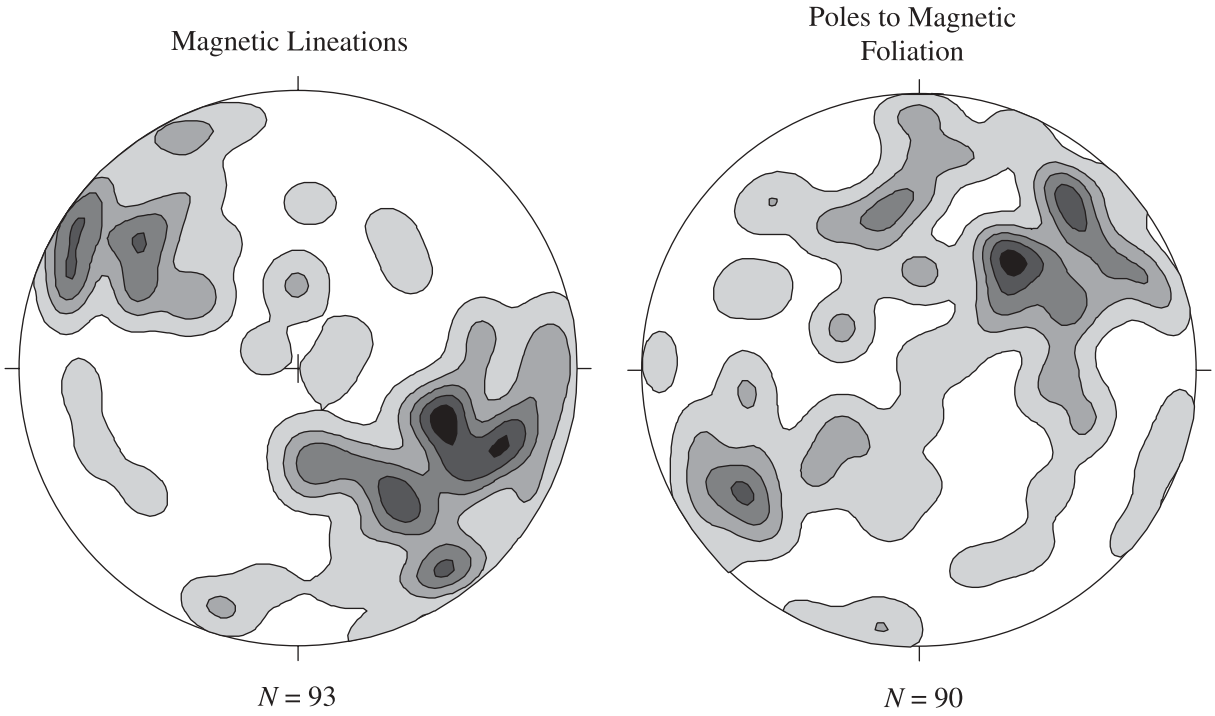


9*b*), the AMS ellispoid shape is triaxial, and both magnetic lineation and magnetic foliation orientations can be obtained. If two of the eigenvalues that characterize AMS have overlapping confidence regions, the AMS ellipsoid has uniaxial symmetry. If, for example, maximum and intermediate eigenvalues overlap, (Fig. 9*d*) the ellipsoid is uniaxial-oblate. Because lineation directions are non-unique for oblate ellipsoids, only the orientation of magnetic foliation can be determined. Similarly, for sites with overlapping confidence regions of intermediate and minimum eigenvalues that have uniaxial prolate elipsoid shapes, only the orientations of magnetic lineations are used. In all, foliation data from 14 sites and lineation data from 11 sites were omitted because the fabrics had uniaxial (or spherical) symmetry. All other sites had triaxial fabrics so both folilation and lineation could be determined.

Figure 10 summarizes the magnetic fabric in the study area by showing lower hemisphere projections for k_{max} and k_{min}. Magnetic lineations trend mostly west-northwest and east-southeast with moderate to shallow plunges; magnetic foliations strike west-northwest with moderately steep southwest dips more common than steep northeast dips. How these fabrics are distributed throughout the area is shown in Fig. 11. The area is divided somewhat arbitrarilly into six different geographic zones to isolate areas with different fabric orientations. Magnetic foliations are plotted in Fig. 11*a* with strike and dip symbols on the map and poles to foliation, k_{min}, on a separate equal area projection for each zone. The magnetic lineations are plotted in Fig. 11*b*. Fabrics are more coherent in some zones than others. For instance, zone F has

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Fig. 10. Contoured lower hemisphere projections of all magnetic fabrics in the Cooper Mountain pluton. The projection on the left is magnetic lineation and the projection on the right shows poles to the magnetic foliation. *N*, number of sites.



fairly uniform southwest dips and northwest plunges, and zone B has steep northeast dips and steep southeast plunges. In contrast, although zone D has fairly uniform southeast plunging lineation, foliation attitudes range widely. These variations between and within zones may reflect local influences of surviving emplacement fabrics or differences in alignment and contributions of the paramagnetic and magnetic phases. Most departures from the common northwest-striking fabric occur in zone A near the northwest pluton margin.

Geochronology

Biotite sample 137a does not form a plateau even though we have obtained a good-flat behavior in the age spectrum (Fig. 12*a*). The calculated isochron age for this sample is 47.16 ± 0.42 Ma ($^{40}Ar/^{36}Ar$ initial = 371.84 ± 51.49; MSWD (means square of weighted deviates) = 0.766; 97.9% of released $^{39}Ar_K$), which agrees with the obtained total gas age at ~47.18 Ma. A second aliquot of biotite for this sample was analyzed by the total fusion technique and yielded a more precise age at 47.65 ± 0.10 Ma. Both ages, the isochron and the total fusion, are the same within limits of analytical error. However, we use the more precise total fusion age as the best estimation at which this plutonic sample cooled below the closure temperature of biotite (300–350 °C).

Biotite sample 235 presents a very similar behavior (Fig. 12*b*) but, in this case, the isochron age is slightly younger at 46.71 ± 0.30 Ma ($^{40}Ar/^{36}Ar$ initial = 320.9 ± 45.38; MSWD = 0.372; 85.1% of released $^{39}Ar_K$), which agrees well with the total gas age of ~46.77 Ma. A second aliquot of biotite yielded a 46.76 ± 0.04 Ma total fusion age. These three ages agree within limits of analytical error, but again we use the more precise total fusion age as the best approximation of the time at which the pluton cooled below the closure temperature of biotite.

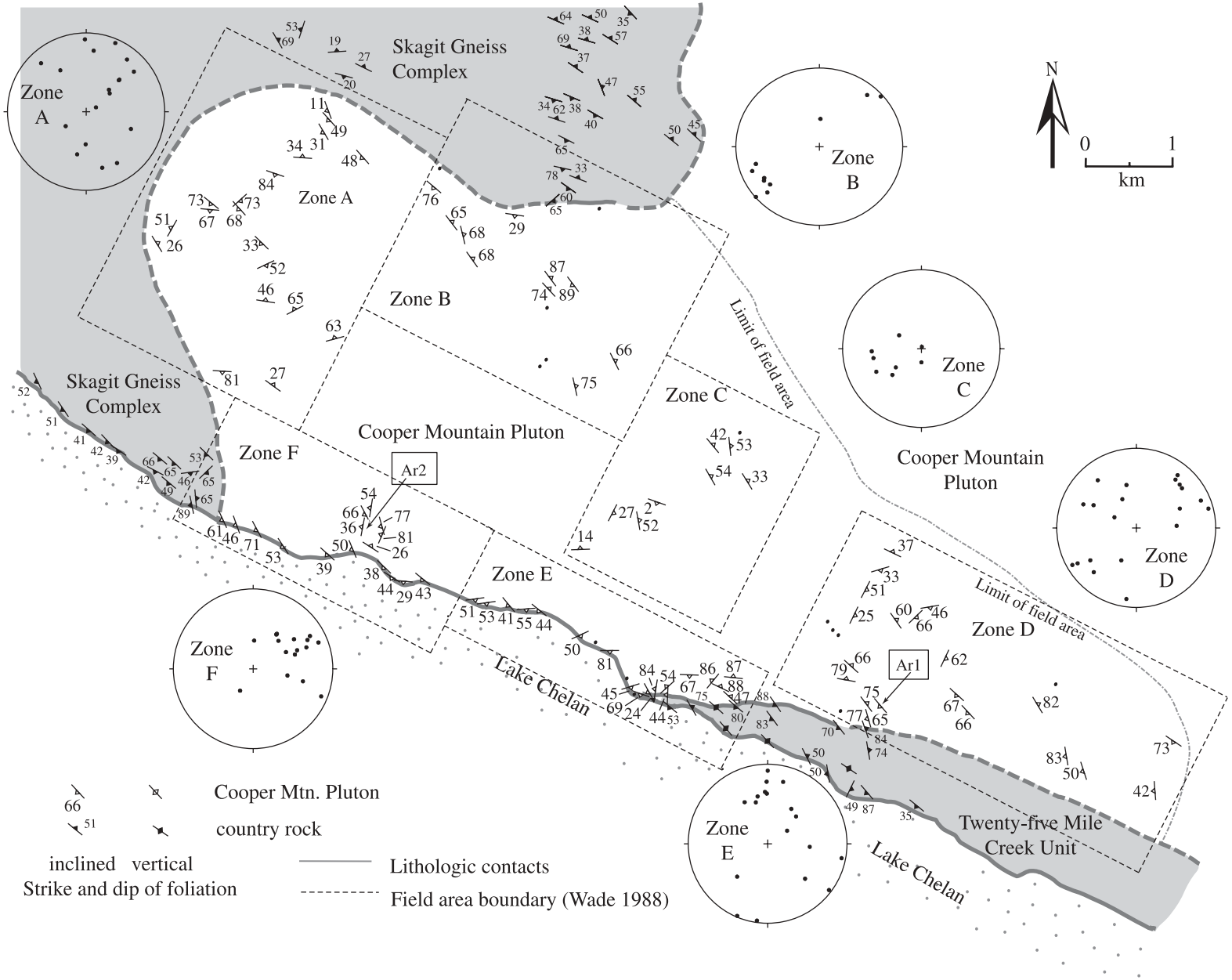
It is important to note that the biotite total fusion ages presented here for the Cooper Mountain pluton samples are not the same within limits of analytical error. This may be indicating a slightly different cooling history for both samples since the emplacement of the pluton. This is a viable possibility because the samples were collected more than 6 km apart in domains F and D (Fig. 11*a*).

Discussion

The ^{40}Ar-^{39}Ar ages of biotite establish that the dated rocks cooled through the nominal Ar closure temperature of biotite (300–350 °C) by ~ 47 Ma. The blocking temperature for remanence carried by magnetite is slightly higher (ca. 500–580 °C); that for pyrrhotite slightly lower (ca. 290–320 °C). This suggests that the magnetization was acquired as the pluton cooled through the argon biotite closure temperature at ca. 47 Ma. This age corresponds to chron 37n, so the normal polarity bias of the CMP remanence can be explained by rapid cooling of most of the body in a normal polarity magnetic field during the Eocene. The coincidence of remanence of the CMP with that expected for its location in North America during the Eocene means that tilting of the Skagit Gneiss Complex (Hopson and Mattinson 1999) was insignificant after 47 Ma. The only motion that the paleomagnetism cannot rule out is rotation of the pluton about an axis that is parallel to the magnetization direction itself. The fabrics discussed henceforth are interpreted as in situ Eocene-aged fabrics, because the similarity of the paleomagnetic direction from the CMP and the expected Eocene magnetic field does not support any significant post-emplacement

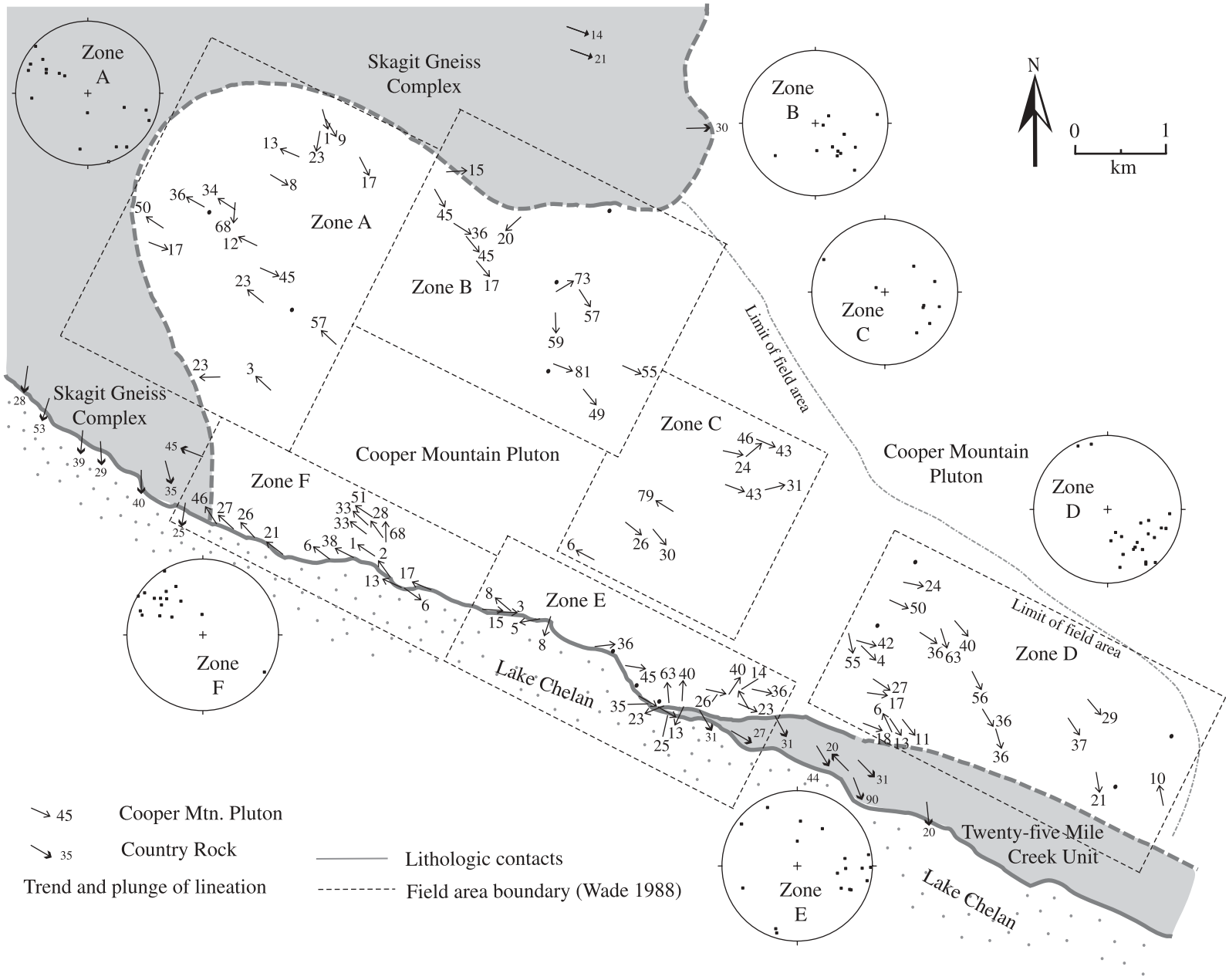
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Fig. 11. Map and equal-area plots of magnetic fabric in the CMP. The field area is broken into six zones; equal-area plots are for zones. (*a*) Map shows magnetic foliation plotted as strike and dip symbols; equal area plots show k_{min} directions (poles to magnetic foliation). (*b*) k_{max} directions (magnetic lineation) plotted. Dots within field area are sites in which foliation or lineation could not be determined. Ar1 and Ar2 locate sites from which ^{40}Ar–^{39}Ar specimen were collected (GC1, sample 137a; GC2, sample 235). Solid symbols show visible fabric measured in the country rock near the contact.



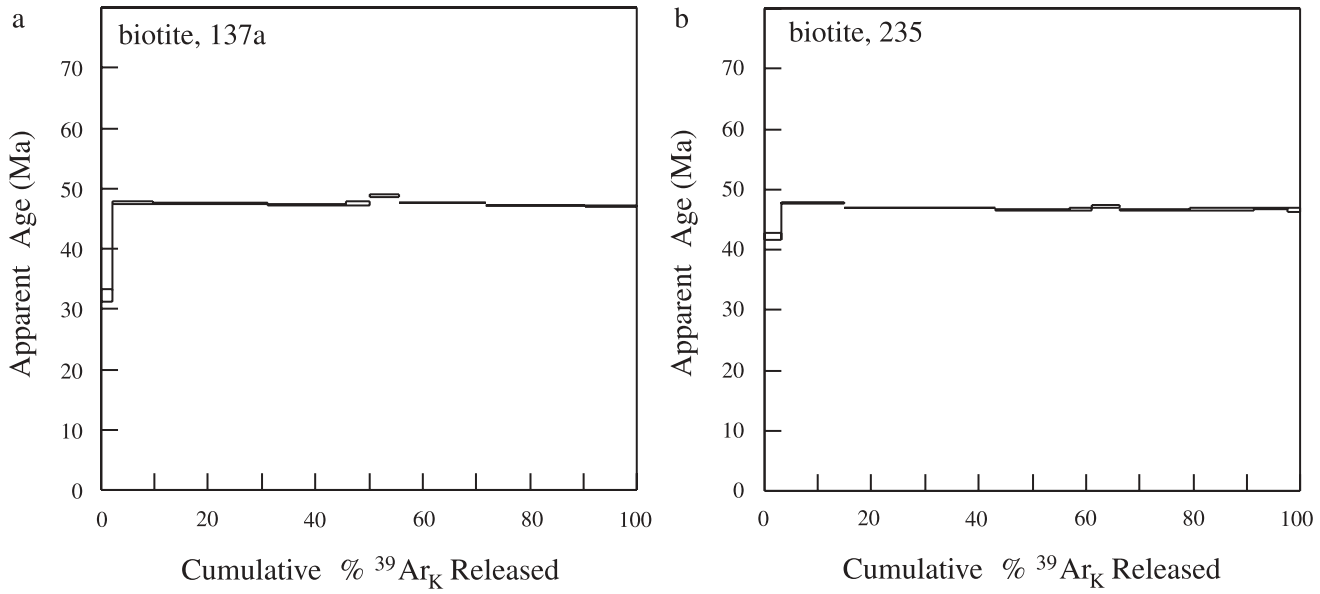
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Fig. 11 (*concluded*).



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Fig. 12. Ar–Ar apparent cooling ages for fractions of ^{39}Ar released during step heating of dated samples. Magnetite, with a slightly higher blocking temperature than argon in biotite, likely has a slightly older remanence but pyrrhotite, with a lower blocking temperature, must have been magnetized slightly later.



reorientation of the pluton. Although it would be interesting to compare these magnetic fabrics with others from the North Cascades, the only other AMS study deals with Cretaceous rocks (Mt. Stuart batholith; Benn et al. 2001), and we are here focusing on interpretations of Eocene tectonics.

The general orientation of the fabric is the first-order feature to interpret. Deviations from the generally steep, southwest-dipping foliation and horizontal, NW–SE-trending lineation arise from several causes, some of which were mentioned in the text sections "Results" and "Anisotropy of magnetic susceptibility"). The average fabric in the pluton is interpreted to result from regional deformation because its overall NW–SE orientation parallels fabric in the country rock and is discordant with the country rock contact. This is perhaps seen best in the lower right of Figs. 11*a* and 11*b*, where the fabrics in the CMP and Twenty-five Mile Creek unit appear parallel, and both are oblique to the contact. Lacking evidence for solid-state deformation in the rocks (see Fig. 2) and the similarity of magnetic and petrofabrics (Fig. 6), the magnetic fabrics shown in Figs. 10 and 11 are interpreted to show tectonic alignment of biotite when the pluton was crystallizing. Therefore, this syntectonic fabric can be used to deduce Eocene-age regional strain.

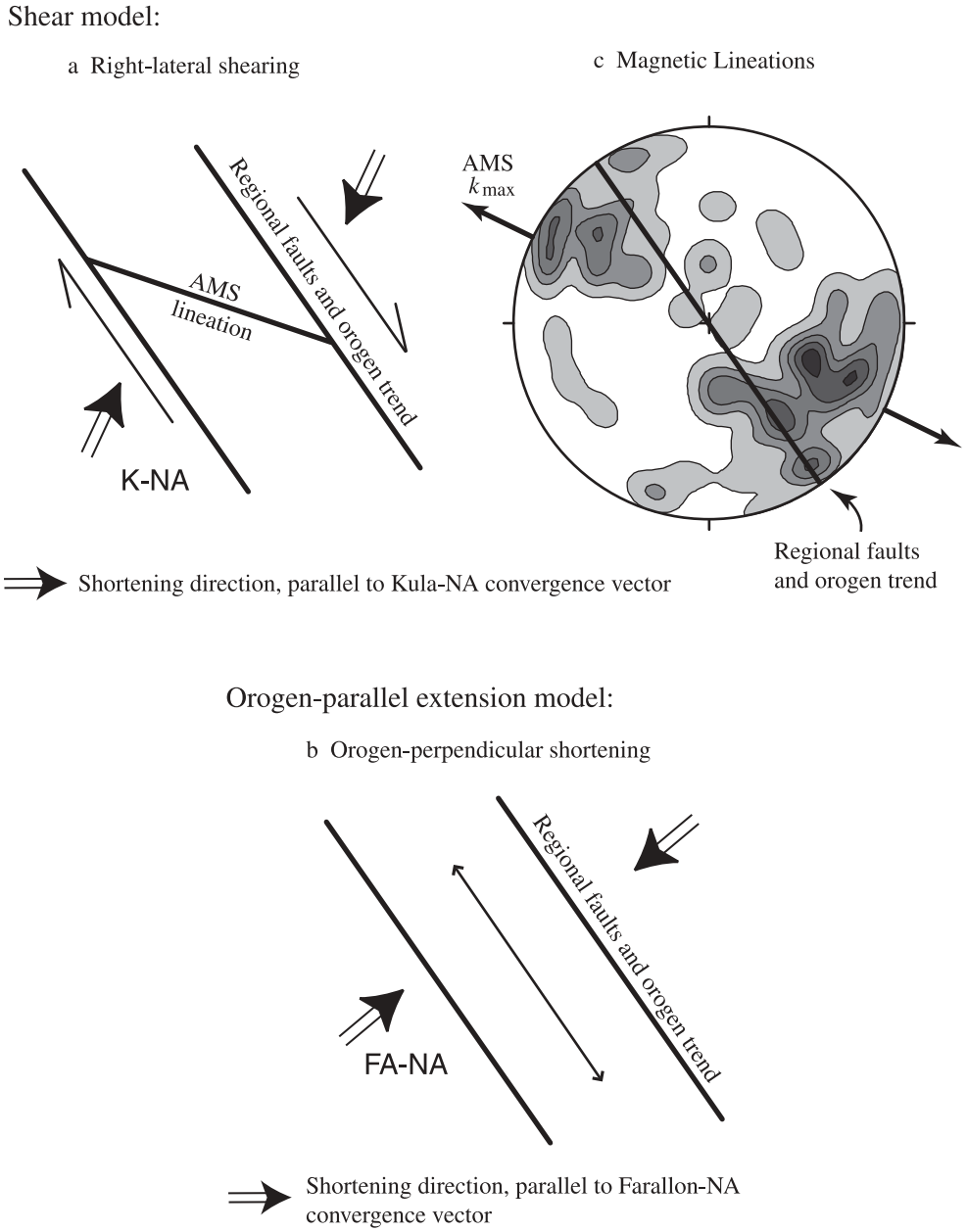
We will consider two mechanisms for generating the fabric. One model to explain fabric orientation in the CMP involves shear, perhaps associated with displacements on the NW–SE faults in the region (i.e., the Ross Lake fault zone, RLFZ) or dextral transpression (Brown and Talbot 1989). The RLFZ is an early Tertiary fault, thought to be active until ~45 Ma, exhibiting dextral strike-slip motion (Misch 1966; Haugerud et al. 1991*b*). The younger age limit of fault movement was based on an interpretation that the Golden Horn batholith and the CMP were undeformed, posttectonic plutons that cut the fault trace and fabrics related to it (Haugerud et al. 1991*b*). However, it has been established here that the CMP is not posttectonic. Simple shear and low-strain transpression can result in horizontal lineations that are oblique to the regional faults and trend of the orogen (Tikoff and Greene 1997), much like the magnetic lineations observed in the CMP. Figure 13*a* is a model of such a system with the lineation direction horizontal and oblique to the trend of the orogen. Figure 13*b* shows the obliquity of the lineations.

The second model is orogen-parallel extension (Fig. 13*c*). Some workers studying rocks in the North Cascades crystalline core have concluded that there was significant orogen-parallel (NW–SE) stretching during the Eocene (Ewing 1980; Miller et al. 2000). The orogen-parallel extension models assume that shortening was more or less orthogonal to the NW–SE trend of the orogen, producing a subhorizontal stretching of the orogen via pure shear. If this strain regime were active during emplacement and cooling of the CMP, the lineation recorded by AMS should also parallel the extension direction. The obliquity (~30°) between the lineation direction in the CMP and both the major trend of the Cascade orogeny and strikes of nearby faults, such as the RLFZ, makes us favor dextral shear strain as the origin of the CMP fabrics.

The dextral shear strain can be further related to the convergence geometry between adjacent tectonic plates (e.g., Tikoff and Tessier 1994; Benn et al. 2001). Using the Diehl et al. (1983) reference pole, the margin of North America in this region was oriented essentially north–south. Based on analysis of regional patterns of magmatism, geochemical variations of the magmas, and incorporation of existing plate motion models and other geological data, Breitsprecher et al. (2003) concluded that the Kula–Farallon ridge was being subducted beneath North America near the latitude of the CMP at 50 Ma. Thus the Eocene strain field in this region could be due to either north-oblique convergence of the Kula plate relative to North America or to essentially orthogonal convergence of the Farallon plate relative to North America. Both of these convergence vectors are shown in Fig. 13. For example, for Kula – North America convergence, a regional

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Fig. 13. Two models used to explain fabric orientation. (*a*) shows right-lateral shearing causing lineation to be oblique to the trend of the orogen and regional faults. K-NA, Kula – North America convergence. (*b*) is a lower-hemisphere contour of Cooper Mountain pluton magnetic lineations showing the average direction of magnetic lineation at an acute angle to the trend of the orogen. (*c*) shows orogen-perpendicular shortening resulting in extension parallel to the orogen. FA-NA, Farallon – North America convergence. AMS, anisotropy of magnetic susceptibility.



shortening direction that is oblique to the regional orogenic trend (Figure 13*a*) would be expected. For Farallon – North America convergence, a regional shortening direction that is essentially orthogonal to the regional orogenic trend is expected (Fig. 13*b*). If the lineation recorded by AMS from the CMP is orthogonal to the direction of regional shortening in the Cascades, it appears to support an oblique-convergence model as in Fig. 13*a*. Therefore, the Eocene strain field at the CMP ca. 47 Ma may be better attributed to convergence of the Kula plate than the Farallon plate relative to North America.

Conclusions

Using AMS we were able to map the subtle fabric present in the CMP. AMS reveals predominant NW–SE strike of foliations with shallow lineations. Coincidence of observed and expected paleomagnetic directions suggests that there was no reorientation of the CMP since the pluton cooled below 550 °C and, therefore, no need to correct the fabrics for reorientation. These in situ fabrics are consistent with a model of Eocene dextral shearing resulting from regional

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strain in the North Cascades crystalline core, and more consistent with strain resulting from North America near the CMP interacting with the Kula plate than with the Farallon plate ca. 47 Ma. AMS and paleomagnetic work on other plutons in the North Cascades, such as the Eocene Golden Horn batholith, can provide additional data to constrain internal deformation of the North Cascades particularly during Eocene events in the Chelan block of the North Cascades crystalline core.

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Appendix A

Table A1. Site locations and sample numbers.

Site No.	Sample No.	E (m)	N (m)	*N*/*n*
2	2	704836	5324448	1/4
3	3	703779	5324402	1/5
4	4	703309	5324230	1/4
10	10	695155	5324848	1/3
11	11	694456	5324687	1/0
12	12	694013	5324762	1/0
48	24	700118	5322658	1/2
49	25	699969	5322774	1/7
50	26	699850	5322886	1/4
52	28	700301	5322707	1/3
56	31	700598	5322822	1/2
57	32	700967	5323057	1/2
60	35	701373	5323012	1/2
61	36	701514	5322882	1/4
62	37	701734	5323105	1/3
63	38	701686	5322740	1/3
72	46	699601	5323370	1/3
73	47	699392	5323552	1/3
74	48a, 48b	699098	5323612	2/6
75	49b	698372	5324014	1/2
76	50	698201	5324002	1/4
80	54	695710	5325550	1/2
81	54a	695863	5324390	1/3
82	55a, 55b	696044	5324345	2/5
83	56a, 56b	696369	5324408	2/6
84	57b	696488	5324360	2/0
85	58	697278	5324099	1/5
86	59	697482	5324073	1/2
95	65	704050	5322591	1/3
97	66	704005	5322811	1/5
101	68	703640	5323202	1/4
102	69	703614	5322681	1/2
112	75	700598	5322688	1/2
114	76	700167	5322688	1/2
122	83	692925	5325175	1/4
123	84	693149	5325131	1/3
124	85	693491	5325067	1/2
125	86a, 86b	693938	5324781	2/2
126	87a	694307	5324714	2/0
127	88	694716	5324717	1/5
128	89	690934	5326303	1/4
176	114	700349	5326124	1/3
178	116	699092	5325747	1/5
179	117	700033	5325747	1/5
181	118	699056	5325170	1/4
182	119b	703528	5324039	1/3
184	120	704366	5325528	1/4
185	121	703435	5324118	1/2
186	122	703081	5324583	1/0
187	123	701252	5327208	1/3
188	124	701224	5326645	1/5
189	125	698385	5328497	1/4
190	126b	698781	5329842	1/4
197	132b	701255	5331064	2/0
199	133a	701848	5326598	1/4
200	134a	701545	5327222	1/3
202	135	703741	5323459	1/1
205	137a	704266	5322811	1/6
209	139	701722	5327445	1/2
213	142	697755	5331005	1/2
214	143	697535	5331176	1/3
224	153b	698375	5331294	2/0
236	157b	699211	5331254	1/4
237	158	699073	5330655	1/0
240	160	698874	5328055	1/5
300	200	693639	5331387	1/3
301	201	693038	5330833	1/3
302	202	693029	5330782	1/2
303	203	692550	5330754	1/2
304	204a, 204b	692554	5330689	1/3
305	205	691880	5330289	1/3
306	206	691656	5330000	1/3
308	208	694062	5331727	1/2
310	209b	694425	5332211	1/5
312	210	694542	5332448	1/2
313	211	694500	5332583	1/3
316	213	695119	5331815	1/3
318	214	693429	5330144	1/3
320	215	693536	5329777	1/3
321	216	693550	5329139	1/1
322	217	692838	5327818	1/5
325	219a	694044	5329028	1/2
327	220	694737	5328553	1/2
332	222	693609	5333983	1/0
344	227	694695	5333500	1/0
345	228	695610	5325235	1/3
346	229	695654	5325164	1/5
347	230	695476	5324903	1/4
353	234	695323	5325283	1/3
354	235	695364	5325477	1/4
355	236	695424	5325588	1/4
358	239	700341	5322767	1/3
360	241	709289	5322559	1/5
362	243	707767	5321921	1/2
363	244	707465	5322154	1/5
364	245	696552	5331694	1/6
365	246	696361	5331378	1/6
366	247	696687	5330782	1/4
367	248	696868	5330587	1/7
368	249	697045	5330172	1/6
370	251	698395	5329921	1/3
371	251	698372	5329754	1/3
372	253	698390	5329353	1/5
373	254	698278	5328367	1/3
374	255	699526	5328567	1/4
375	256	703970	5324872	1/6
376	257	704175	5325221	1/0
377	258	704477	5324341	1/7
378	259	705059	5324597	1/4
379	260	705361	5323746	1/6
380	261	705580	5323085	1/2
381	262	705790	5322796	1/6
382	263	706953	5323034	1/4
383	264	707288	5323424	1/3
384	265d	708991	5321661	1/2

Note: Sample numbers refer to block samples collected in the field and drilled in the lab. E and N are, respectively, Universal Transverse Mercator (UTM) Zone 10, Easting and Northing coordinates of site locations. *N*, number of block samples per site; *n*, specimens per site.

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Table A2. $^{40}Ar/^{39}Ar$ step-heating data.

Step	*T* (°C)	$\%^{39}Ar$ of total	Radiogenic yield (%)	$^{39}Ar_k$ (Mol × 10^{-12})	$\frac{^{40}Ar^*}{^{39}Ar_k}$	Apparent K/Ca	Apparent K/Cl	Apparent Age (Ma)	Error (Ma)
137a Cooper Mt. biotite, *J* = 0.004754 ± 0.50%, weight = 19.2 mg, #31KD22									
A	700	2.1	58.3	0.046587	3.806	55.7	163	32.35	± 0.53
B	800	7.8	91.8	0.174909	5.634	137.2	363	47.68	± 0.10
C	900	21.3	98.4	0.479084	5.627	148.6	476	47.63	± 0.06
D	1000	14.5	97.7	0.325568	5.582	50.1	461	47.25	± 0.04
E	1050	4.4	94.5	0.099534	5.613	42.8	368	47.51	± 0.15
F	1100	5.6	95.2	0.125143	5.769	81.5	399	48.81	± 0.09
G	1150	16.0	98.6	0.358728	5.624	200.6	473	47.60	± 0.04
H	1200	18.7	99.0	0.420405	5.571	134.3	465	47.16	± 0.02
I	1250	9.7	98.4	0.216818	5.562	86.2	73	47.08	± 0.05
Total Gas		100.0	96.8	2.246776	5.573	122.7	408	47.18	
137a Cooper Mt. biotite, TF, *J* = 0.004748 ± 0.50%, weight = 2.3 mg, #30KD22									
A	1450	100.0	95.0	0.258194	5.637	72.9	74	47.65	± 0.10
235 Cooper Mt. biotite, *J* = 0.004745 ± 0.50%, weight = 23.4 mg, #33KD22									
A	700	3.2	73.3	0.101583	4.989	42.5	163	42.21	± 0.27
B	800	11.8	96.0	0.379051	5.650	298.0	211	47.72	± 0.07
C	900	28.1	99.0	0.902834	5.555	616.4	219	46.94	± 0.04
D	1000	13.9	98.3	0.446623	5.525	270.6	214	46.68	± 0.05
E	1050	4.1	96.1	0.131491	5.536	85.9	204	46.78	± 0.10
F	1100	5.2	96.6	0.166983	5.576	76.1	207	47.11	± 0.11
G	1150	13.1	98.6	0.421478	5.521	117.6	218	46.65	± 0.04
H	1200	12.1	98.7	0.389072	5.530	124.7	215	46.72	± 0.07
I	1250	6.1	95.3	0.195178	5.544	137.3	56	46.84	± 0.10
J	1350	2.5	97.5	0.081778	5.521	97.4	187	46.65	± 0.17
Total Gas		100.0	97.1	3.216071	5.535	295.8	203	46.77	
235 Cooper Mt. biotite, TF, *J* = 0.004752 ± 0.50%, weight = 3.4 mg, #32KD22									
A	1450	100.0	96.7	0.492557	5.526	180.8	110	46.76	± 0.04

Note: Ages calculated assuming an initial $^{40}Ar/^{36}Ar = 295.5 \pm 0$. All precision estimates are at the one sigma level of precision. Ages of individual steps do not include error in the irradiation parameter *J*. No error is calculated for the total gas age. TF denotes total fusion experiment.

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Table A3. Paleomagnetic directions.

Site No.	Specimen	*D* (°)	*I* (°)	MAD (°)	Begin *T* (°C)	End *T* (°C)
Pyrrhotite components						
2	2-c2-a	335.8	71.8	4.7	150	330
3	3-c1-d	86.7	76.7	5.1	250	340
49	25-c2-b	313.8	52.3	7.1	150	370
52	28-c1-a	64.0	83.6	3.4	90	340
57	32-c1-b	336.7	56.5	6.2	320	350
60	35-c1-b	329.0	50.5	6.0	290	340
61	36-c2-c	339.4	66.8	1.9	150	340
62	37-c2-b	165.8	–62.9	4.6	150	320
72	46-c1-b	5.9	66.2	5.1	325	370
95	65-c1-c	343.4	72.4	0.8	250	355
101	68-c1-b	358.9	44.0	6.5	260	340
114	76-c1-c	17.0	80.9	5.3	150	370
122	83-c2-c	35.3	66.4	4.8	290	330
123	84-c1-d	21.8	82.8	3.5	150	370
176	114-c1-c	340.0	68.9	4.3	310	350
178	116-c2-a	24.2	70.0	0.6	260	350
184	120-c2-a	311.3	70.7	6.5	260	340
185	121-c1-b	354.6	79.7	2.5	300	340
199	133a-c1-b	252.2	82.4	7.1	100	300
202	135a-c1-b	54.8	51.0	0.5	225	370
205	137a-c1-c	9.7	80.5	1.6	210	350
209	139-c2-b	4.0	76.2	1.6	275	340
213	142-c3-b	159.7	–21.8	2.0	300	340
240	160-c2-a	311.6	53.4	7.6	290	350
300	200-c1-c	318.6	71.5	6.5	150	340
301	201-c1-a	23.0	53.7	1.9	150	370
305	205-c1-b	316.7	72.9	7.5	290	350
313	211-c1-c	129.9	–29.6	2.3	150	350
316	213-c1-a	130.1	–3.6	6.3	150	370
318	214-c2-b	281.5	32.9	6.8	250	400
322	217-c2-c	60.4	50.5	7.3	260	350
327	220-c1-b	100.1	26.9	7.4	225	370
346	229-c1-a	53.6	63.5	5.1	250	340
353	234-c2-c	344.7	68.5	1.1	330	360
354	235-c2-c	203.9	–0.9	5.8	250	325
358	239-c2-b	256.5	52.4	2.6	150	370
366	247-c1-b	322.5	58.3	5.3	250	400
367	248-c2-b	86.7	80.4	5.1	100	275
368	249-c2-a	323.4	75.4	2.8	150	370
370	251-c1-b	308.0	68.3	6.6	320	450
371	252-c1-b	222.7	81.5	7.9	150	290
372	253-c3-b	18.5	74.4	2.5	150	370
373	254-c2-b	235.0	70.9	5.0	290	450
375	256-c1-d	45.5	–14.5	7.7	330	360
378	259-c2-b	335.8	55.9	5.2	290	340
380	261-c1-b	325.8	72.3	3.7	150	350
381	262-c1-c	321.3	50.6	2.4	150	370
382	263-c2-a	305.1	35.9	2.6	150	370
Magnetite components						
4	004-c1-d	269.2	57.5	7.7	390	525
50	026-c2-b	343.7	62.7	2.9	300	565
52	028-c2-b	38.3	71.3	4.9	370	600
95	065-c2-a	351.0	71.1	6.9	370	575
102	069-c2-a	36.1	59.9	1.4	370	600
114	076-c1-c	336.7	66.0	7.1	370	585

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Table A3 (*concluded*).

Site No.	Specimen	*D* (°)	*I* (°)	MAD (°)	Begin *T* (°C)	End *T* (°C)
187	123-c2-c	318.8	75.4	2.6	340	580
188	124-c2-b	310.7	51.2	5.9	400	580
199	133a-c2-a	290.4	50.8	6.6	300	565
209	139-c1-b	58.4	76.8	5.5	400	565
301	201-c2-b	24.4	61.9	0.6	300	580
305	205-c3-b	39.6	75.5	1.7	510	575
305	205-c3-b	222.6	2.8	4.0	400	510
308	208-c2-a	43.1	69.4	1.5	150	535
355	236-c1-b	42.7	72.3	5.3	100	600
360	241-c3-b	178.7	–28.7	7.8	620	665
370	251-c1-b	308.0	68.3	6.6	320	450
379	260-c2-b	298.4	61.4	5.0	370	565

Note: *D* and *I* are declination and inclination, respectively, of free lines fit using principal component analysis (PCA) (Kirschvink 1980) to Npoints along demagnetization path over temperature range Begin *T* to End *T*. MAD is maximum angular deviation of line (°) (Kirschvink 1980). Directions are in situ.

Table A4. Site Mean AMS and Susceptibility.

		k_{max}		k_{min}		Eigenvalues (normalized)			
Site No.	*M*	*D* (°)	*I* (°)	*D* (°)	*I* (°)	Max	Int	Min	Bulk *k* (SI)
2	4	161.4	63.1	316.1	24.6	0.3466	0.3301	0.3233	7.28E–05
3	5			289.7	64.9	0.3374	0.3365	0.3261	9.42E–05
4	4	174.7	54.8			0.3379	0.3322	0.3299	6.43E–05
10	3	306.4	32.3	67.2	39.1	0.3438	0.3337	0.3226	1.12E–04
48	2					0.3412	0.3303	0.3285	5.10E–05
49	7	119.5	35.9	341.1	46.0	0.3369	0.3337	0.3293	1.06E–04
50	4					0.3359	0.3338	0.3303	1.70E–04
52	3	246.7	23.5	53.5	65.9	0.3429	0.3331	0.3241	4.68E–05
56	2	3.7	40.4	133.2	36.7	0.3386	0.3349	0.3266	5.97E–05
57	2	102.5	23.6	359.7	26.9	0.3403	0.3321	0.3276	7.06E–05
60	2	33.8	39.3	127.1	4.0	0.3374	0.3337	0.3290	7.89E–05
61	4	114.0	23.7	205.6	3.5	0.3385	0.3320	0.3295	8.74E–05
62	3	98.6	36.7	190.5	2.6	0.3393	0.3320	0.3287	1.02E–04
63	3	333.2	17.4	226.7	42.4	0.3393	0.3350	0.3258	1.27E–04
72	3	99.5	45.7	359.8	9.3	0.3409	0.3335	0.3257	1.14E–04
73	3	61.6	33.3			0.3402	0.3303	0.3295	7.57E–05
74	6	86.3	37.0	315.1	41.2	0.3391	0.3337	0.3272	6.99E–05
75	2	294.8	8.2	33.2	45.6	0.3396	0.3328	0.3277	1.01E–04
76	4	81.7	3.0	348.1	49.6	0.3397	0.3318	0.3284	9.40E–05
80	2			62.1	52.9	0.3423	0.3356	0.3221	1.60E–04
81	3	291.0	12.7	34.4	45.7	0.3416	0.3329	0.3255	1.03E–04
82	5	121.0	6.2	17.9	64.3	0.3399	0.3317	0.3285	8.59E–05
83	6	287.9	16.4	36.0	46.5	0.3398	0.3335	0.3267	8.43E–05
85	5	91.6	15.8	348.2	39.4	0.3406	0.3343	0.3252	1.14E–04
86	2	91.4	3.2	358.9	38.2	0.3408	0.3335	0.3257	1.04E–04
95	3	335.3	6.6	66.8	13.3	0.3424	0.3362	0.3214	1.00E–04
97	5	144.3	13.9	238.2	15.0	0.3417	0.3321	0.3262	1.34E–04
101	4	94.2	17.1	188.0	12.1	0.3555	0.3315	0.3131	1.72E–05
102	2	106.5	19.9			0.3356	0.3330	0.3315	1.42E–04
112	2	199.0	13.5	94.3	46.6	0.3387	0.3338	0.3275	6.30E–05
114	2	113.9	25.0	13.4	21.2	0.3378	0.3336	0.3287	1.03E–04
122	4	327.2	18.5	67.7	28.6	0.3427	0.3317	0.3257	6.91E–05
123	3	311.1	27.3	71.1	44.1	0.3515	0.3301	0.3185	6.08E–05
124	2	319.5	25.6	59.4	19.7	0.3394	0.3344	0.3262	2.10E–04
125	3	306.8	21.0	53.2	36.3	0.3532	0.3368	0.3100	7.16E–05
127	5			43.5	52.7	0.3404	0.3341	0.3255	1.09E–04
128	4	323.6	1.2	54.0	16.1	0.3404	0.3353	0.3244	1.21E–04

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Table A4 (*concluded*).

Site No.	*M*	k_{max}		k_{min}		Eigenvalues (normalized)			
		D (°)	*I* (°)	*D* (°)	*I* (°)	Max	Int	Min	Bulk *k* (SI)
176	3	254.7	87.4			0.3401	0.3318	0.3281	6.73E–05
178	5	126.4	25.7	301.5	64.2	0.3446	0.3343	0.3211	6.87E–05
179	5	141.9	30.3	258.9	37.8	0.3450	0.3327	0.3223	8.01E–05
181	4	298.3	6.0	182.2	76.6	0.3360	0.3340	0.3301	9.51E–05
182	3	137.3	4.6			0.3419	0.3297	0.3284	8.18E–05
184	4			215.3	59.5	0.3363	0.3343	0.3294	3.55E–05
185	2	104.8	41.6			0.3412	0.3304	0.3284	8.57E–05
187	3	105.5	24.4	227.1	49.1	0.3362	0.3341	0.3297	8.09E–05
188	5	105.8	45.1	243.6	36.4	0.3374	0.3344	0.3282	1.29E–04
189	4	104.9	80.5			0.3360	0.3324	0.3317	1.27E–04
190	4	143.7	57.0	48.0	3.7	0.3460	0.3346	0.3195	1.40E–04
199	4	73.0	31.0	235.3	57.8	0.3403	0.3330	0.3268	7.87E–05
200	3	48.3	45.4	268.2	37.1	0.3376	0.3333	0.3291	7.58E–05
202	1	121.0	27.0	224.0	24.0	0.3477	0.3291	0.3232	1.01E–04
205	6	144.3	11.5	48.8	25.4	0.3475	0.3318	0.3207	7.19E–05
209	2	123.6	46.6			0.3436	0.3313	0.3251	8.92E–05
213	2	231.2	19.2	1.5	61.7	0.3467	0.3369	0.3164	1.34E–04
214	3	151.9	37.0	243.0	1.4	0.3471	0.3301	0.3228	1.66E–04
236	4					0.3363	0.3325	0.3312	1.19E–04
240	5	142.5	47.6	249.4	14.9	0.3377	0.3340	0.3284	1.09E–04
300	3	114.3	8.2	23.2	7.4	0.3370	0.3339	0.3292	8.80E–05
301	3	177.8	67.5	316.4	17.3	0.3448	0.3320	0.3232	1.49E–04
302	2	300.4	34.0	43.0	17.9	0.3416	0.3302	0.3283	1.39E–04
303	2	295.7	35.5	38.2	16.8	0.3437	0.3332	0.3232	7.40E–05
304	3	104.8	17.9	6.8	23.3	0.3420	0.3370	0.3211	8.87E–05
305	3	307.0	51.4	114.1	37.9	0.3432	0.3332	0.3236	2.14E–04
306	3	106.2	17.3	235.4	63.8	0.3466	0.3356	0.3178	1.02E–04
308	2	291.0	12.0	182.4	56.3	0.3555	0.3298	0.3147	2.76E–04
310	5	187.1	23.9	51.7	58.1	0.3614	0.3396	0.2991	6.29E–05
312	2	143.6	8.7	45.7	42.0	0.3476	0.3317	0.3208	1.46E–04
313	3			67.3	78.4	0.3492	0.3317	0.3191	3.86E–05
316	3	153.0	17.8	46.6	41.5	0.3572	0.3395	0.3033	6.02E–05
318	3	297.0	11.5	45.4	57.2	0.3411	0.3332	0.3256	2.09E–04
320	3	110.0	45.4	328.1	37.8	0.3445	0.3348	0.3207	1.75E–04
321	1	307.0	23.0	192.0	44.0	0.3404	0.3342	0.3254	1.19E–04
322	5	267.5	26.8	2.8	10.4	0.3541	0.3384	0.3075	2.16E–04
325	2	5.4	59.7	148.6	25.0	0.3451	0.3371	0.3178	7.68E–05
327	2	306.1	56.5	164.1	27.5	0.3380	0.3330	0.3290	1.16E–04
345	3	331.1	27.7	68.2	13.2	0.3398	0.3328	0.3273	8.73E–05
346	5	358.6	67.6	110.2	8.7	0.3414	0.3332	0.3254	7.59E–05
347	4			208.4	65.2	0.3393	0.3340	0.3267	7.70E–05
353	3	305.1	32.9	101.8	54.8	0.3415	0.3330	0.3254	1.00E–04
354	4	315.1	33.0	62.0	24.0	0.3463	0.3316	0.3222	8.77E–05
355	4	295.3	52.8	104.2	36.6	0.3390	0.3340	0.3271	8.87E–05
358	3	359.7	62.5	101.0	5.8	0.3405	0.3347	0.3248	7.84E–05
360	5	303.5	16.9	40.1	20.6	0.3550	0.3456	0.2994	6.64E–04
362	2			66.0	29.9	0.3344	0.3333	0.3323	2.21E–05
363	5	168.3	21.9	75.3	7.2	0.3435	0.3339	0.3227	8.49E–05
364	6	78.8	14.3			0.3451	0.3291	0.3259	1.84E–04
365	6	146.1	45.3	41.1	14.3	0.3439	0.3311	0.3250	8.24E–05
366	4	120.5	35.6	230.5	25.4	0.3441	0.3330	0.3230	9.75E–05
367	7	139.1	45.3	253.4	22.1	0.3448	0.3345	0.3207	9.78E–05
368	6	140.3	17.4	237.5	21.9	0.3428	0.3337	0.3236	1.79E–04
370	3	304.6	63.5	42.0	3.7	0.3402	0.3367	0.3231	1.07E–04
371	3	62.3	73.9	227.1	15.6	0.3404	0.3369	0.3227	1.02E–04
372	5	182.4	62.3			0.3400	0.3323	0.3277	1.13E–04

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Table A4 (*concluded*).

Site No.	*M*	k_{max}		k_{min}		Eigenvalues (normalized)			Bulk *k* (SI)
		D (°)	*I* (°)	*D* (°)	*I* (°)	Max	Int	Min	
373	3					0.3345	0.3338	0.3317	1.27E–04
374	4	110.8	54.7	239.1	23.7	0.3371	0.3335	0.3294	1.15E–04
375	6	112.5	50.5	296.8	39.4	0.3401	0.3352	0.3247	7.92E–05
377	7	120.8	35.5	234.8	29.6	0.3440	0.3321	0.3240	7.46E–05
378	4	138.2	41.7	344.0	45.3	0.3407	0.3317	0.3276	3.10E–04
379	6	153.1	55.9	294.1	27.8	0.3439	0.3306	0.3255	1.15E–04
380	2	146.7	36.1	38.1	23.6	0.3432	0.3305	0.3263	1.06E–04
381	6	161.0	44.2	45.3	24.1	0.3456	0.3321	0.3222	9.68E–05
382	4	144.5	39.3	241.4	8.4	0.3420	0.3331	0.3249	9.49E–05
383	3	139.5	29.3	235.1	9.9	0.3431	0.3290	0.3279	6.40E–05
384	2	346.0	9.3	86.4	48.0	0.3407	0.3348	0.3245	9.64E–05

Note: *M*, number of specimen used from block samples. k_{max}, k_{min} list declination (*D*) and inclination (*I*) of directions of maximum and minimum susceptibilty. Uniaxial oblate, uniaxial prolate, or spherical symmetry fabrics lack the first, second, or both *D*–*I* pairs. Eigenvalues are for maximum (Max), intermediate (Int), and minimum (Min) susceptibility. Bulk *k*, average bulk susceptibility for all specimens used in a site.

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